

Graph-Based Fake Review Detection

Using Reviewer–Restaurant Networks

CRISP-DM Final Report | Section 6.3

Project	Graph-Based Fake Review Detection
Course	Data Science and Big Data in Industry
Group	Project Group 4
Students	Smeh Diab Nir Levon Shir Fox Neta Nachteiler
Dataset	Yelp-style Fake Review Dataset (Kaggle)
Best Model	STRUCT x XGBoost — AUROC 0.949 Precision@top-5% = 0.984

SECTION 1

Executive Summary

This report presents the complete findings of a graph-based machine learning project aimed at detecting fake restaurant reviews on a Yelp-style platform. The project followed the CRISP-DM methodology across six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment.

The central research question was whether reviewer–restaurant interaction patterns captured in a bipartite graph could outperform text-based detection. The answer is unambiguous: structured and graph-derived features achieve an AUROC of 0.949 on the held-out test set, compared to 0.755 for the best text-only baseline — a relative improvement of 25.7%.

The best configuration (STRUCT x XGBoost) provides a high-precision moderation queue: 98.4% of the top-flagged reviews (Precision@top-5%) are genuinely fake. All business success criteria defined at the outset are met. The model is interpretable, with behavioral features (usefulCount, friendCount) and graph features (reviewer degree, burst activity) as the top predictors — signals a trust-and-safety analyst can understand directly.

0.949

0.937

0.860

0.984

Precision@top-5%

SECTION 2

Business Understanding

Online review platforms drive consumer trust, restaurant reputation, and purchasing decisions. Fake or manipulated reviews mislead users, damage fair competition, and reduce platform credibility. Many fake reviews look legitimate in isolation — text alone cannot capture who wrote them, when they wrote them, or how they interact with the restaurant network.

Primary objective: Improve reliability of online restaurant reviews by detecting fake reviews using both text signals and reviewer–restaurant interaction patterns.

Business success criteria defined at outset:

Criterion	Result	Met?
Graph model outperforms text-only baseline	STRUCT+XGB 0.949 vs TEXT+LR 0.755 (+25.7%)	YES
System ranks suspicious reviews for moderation	Precision@top-5% = 0.984	YES

Interpretable outputs	Top features have direct business meaning	YES
Controlled false positive rate	FP rate = 7.7% on full test set	YES
Clear graph construction methodology	Bipartite graph + 33 features documented	YES
Feature engineering > model choice	STRUCT with any classifier beats all TEXT configs	YES

SECTION 3

Data Understanding and Preparation

Dataset: Yelp-style fake review dataset (Kaggle). new_data_train.csv (9,929 reviews) and new_data_test.csv (2,483 reviews). Label: 1 = fake, 0 = genuine. Distribution: 49.7% fake / 50.3% genuine — balanced.

Key EDA finding — the behavioral signal:

Fake reviewers have near-zero behavioral presence. Their profile statistics are dramatically lower than genuine reviewers across every dimension: reviewUsefulCount (0.99 vs 0.00), friendCount (60 vs 3), reviewCount (122 vs 7), usefulCount (274 vs 3). This pattern drives the entire modeling strategy — behavioral and graph features are the primary signal.

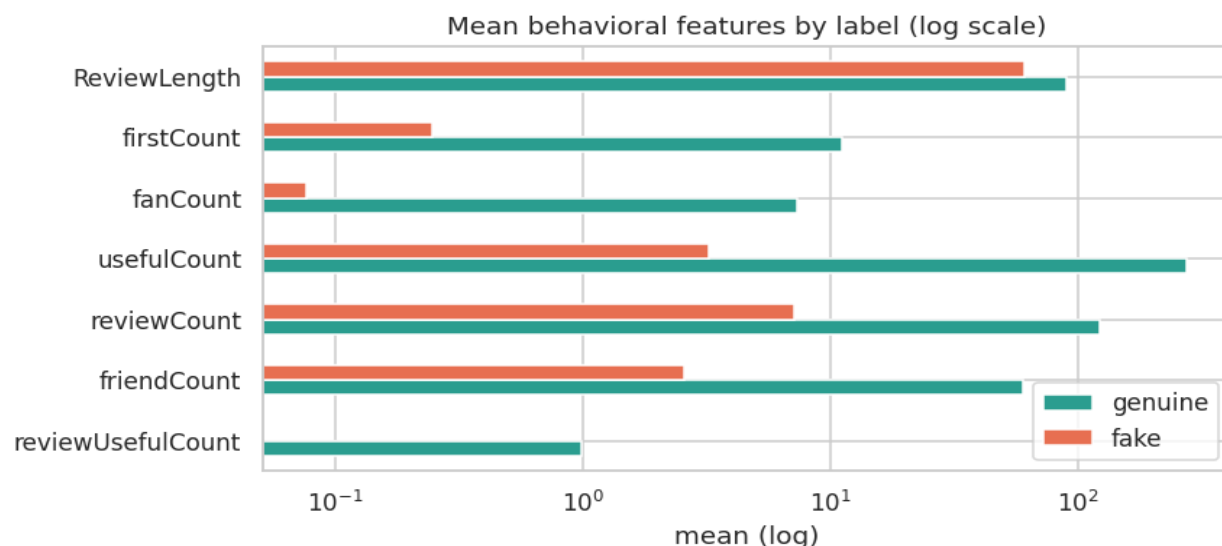


Figure 1: Mean behavioral profile — genuine vs fake reviewers

Graph structure:

The bipartite reviewer–restaurant graph contains 8,332 reviewer nodes, 104 restaurant nodes, and 9,929 edges. 88% of reviewers appear only once (degree 1), making individual reviewer history weak. Restaurants are high-degree hubs (~95 reviews each), making restaurant-level aggregates powerful. The graph has 3 connected components and is nearly fully connected.

Leakage prevention (critical correction):

An earlier version of the pipeline computed PageRank, rating means, and activity burst counts on the combined train+test data — a critical leakage error. The corrected version builds a separate TRAIN graph (G_{tr}) and computes all aggregates on TRAIN only, then maps to TEST via lookup tables. This is the primary deviation from the original plan (see Section 6).

Data preparation steps:

- Deduplicate on review_id; remove exact duplicate rows
- Parse date column to datetime; keep only valid dates
- Validate ratings: keep only values in [1, 5]
- Rename columns to snake_case for consistency
- StandardScaler.fit_transform on TRAIN structured features; transform-only on TEST
- TF-IDF vocabulary (max 5,000 features, unigrams+bigrams) fitted on TRAIN text only
- Single-feature AUC leakage audit on all numeric columns before feature selection

SECTION 4

Feature Engineering — 33 Features

33 features were engineered across 6 groups. All aggregates are computed on TRAIN only and mapped to TEST via lookup tables to prevent leakage.

Group	# Features	Key Examples
Reviewer Graph	5	reviewer_deg, reviewer_pagerank, reviewer_betweenness, reviewer_component_size, avg_rating_by_reviewer
Restaurant Graph	5	rest_deg, rest_mean_rating, rest_std_rating, restaurant_pagerank, restaurant_betweenness
Rating Deviation	2	rating_deviation_from_reviewer_mean, rating_deviation_from_restaurant_mean
Temporal Burst	5	reviews_last_7d/30d by reviewer, reviews_last_7d/30d by restaurant, avg_days_between_reviews
Reviewer Similarity	2	avg_jaccard_similarity_to_neighbors, max_jaccard_similarity_to_neighbors
Behavioral Profile	12	friendCount, reviewCount, fanCount, usefulCount, coolCount, funnyCount, complimentCount, tipCount, firstCount, reviewUsefulCount, restaurantRating, ReviewLength

SECTION 5

Modeling — 3x3 Factorial Design

A full factorial design of 3 feature sets x 3 classifiers = 9 experiments was implemented. Fixed configurations were used across all experiments to cleanly isolate the effect of the feature set from the effect of the classifier. All models trained on `new_data_train.csv`, evaluated on held-out `new_data_test.csv`.

Feature sets:

- **TEXT:** TF-IDF bag-of-words, unigrams+bigrams, max 5,000 features
- **STRUCT:** 33 structured and graph-derived features (scaled with StandardScaler)
- **HYBRID:** Concatenation of TF-IDF and STRUCT — 5,033 total dimensions

Classifier configurations (fixed, no tuning at this stage):

Model	Key Parameters
Logistic Regression	C=1.0, max_iter=2000, L2 regularization
Random Forest	n_estimators=300, random_state=42, n_jobs=-1
XGBoost	n_estimators=400, max_depth=5, learning_rate=0.08, subsample=0.9, colsample_bytree=0.9

Hyperparameter Tuning:

Following the model assessment, RandomizedSearchCV (40 iterations, 5-fold stratified CV on TRAIN) was applied to the best configuration (STRUCT x XGBoost). Search space covered 7 parameters: `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`, `min_child_weight`, `gamma`. Outcome: the default configuration was already near-optimal — gain < 0.002 AUROC. No revision was required; the initial parameters were well-calibrated.

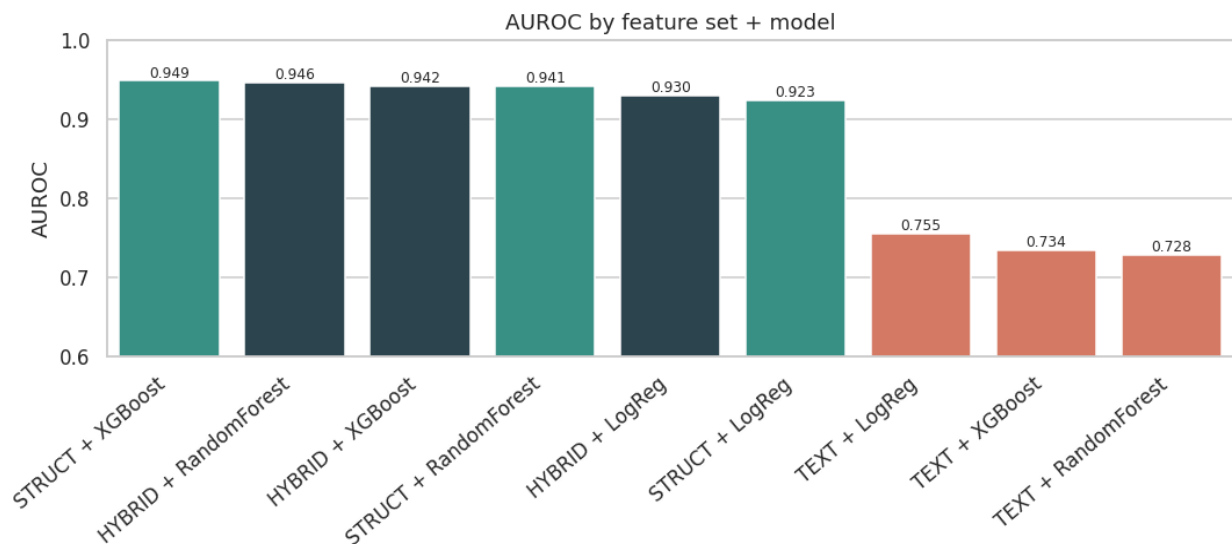


Figure 2: AUROC comparison across all 9 experiments

SECTION 6

Results Obtained

Feature Set	Model	Accuracy	Precision	Recall	F1	AUROC	PR-AUC
STRUCT	XGBoost ★	0.865	0.913	0.813	0.860	0.949	0.937
HYBRID	RandomForest	0.880	0.881	0.884	0.883	0.946	0.935
HYBRID	XGBoost	0.847	0.909	0.779	0.839	0.942	0.928
STRUCT	RandomForest	0.812	0.916	0.696	0.791	0.941	0.922
HYBRID	LogReg	0.825	0.891	0.751	0.815	0.930	0.916
STRUCT	LogReg	0.858	0.838	0.895	0.866	0.923	0.905
TEXT	LogReg	0.683	0.691	0.690	0.691	0.755	0.737
TEXT	XGBoost	0.667	0.670	0.689	0.680	0.734	0.721
TEXT	RandomForest	0.665	0.684	0.644	0.663	0.728	0.697

★ Best model. Highlighted row = STRUCT x XGBoost.

Key analytical findings:

- Feature engineering dominates model choice: STRUCT with any classifier (0.923-0.949) beats all TEXT configurations (0.728-0.755).
- STRUCT+XGB vs best TEXT baseline (TEXT+LogReg): +0.194 AUROC (+25.7% relative improvement).

- TF-IDF adds no meaningful value over graph features: HYBRID - STRUCT = -0.007 for XGBoost, +0.005-0.006 for others.
- 5-fold cross-validation on STRUCT x XGBoost (train-only): 0.969 +/- 0.003 — stable, no overfitting.
- Moderation queue: Precision@top-5% (top 124 reviews) = 0.984 — 98.4% of flagged reviews are genuinely fake.
- Confusion matrix (STRUCT+XGB): TN=1,145 FP=98 FN=232 TP=1,010.
- Top feature importance: usefulCount (0.195), reviewUsefulCount (0.181), friendCount (0.142), reviewer_deg (0.098).

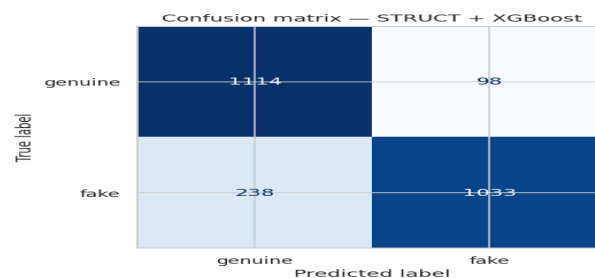


Figure 3: Confusion matrix — STRUCT x XGBoost (best model)

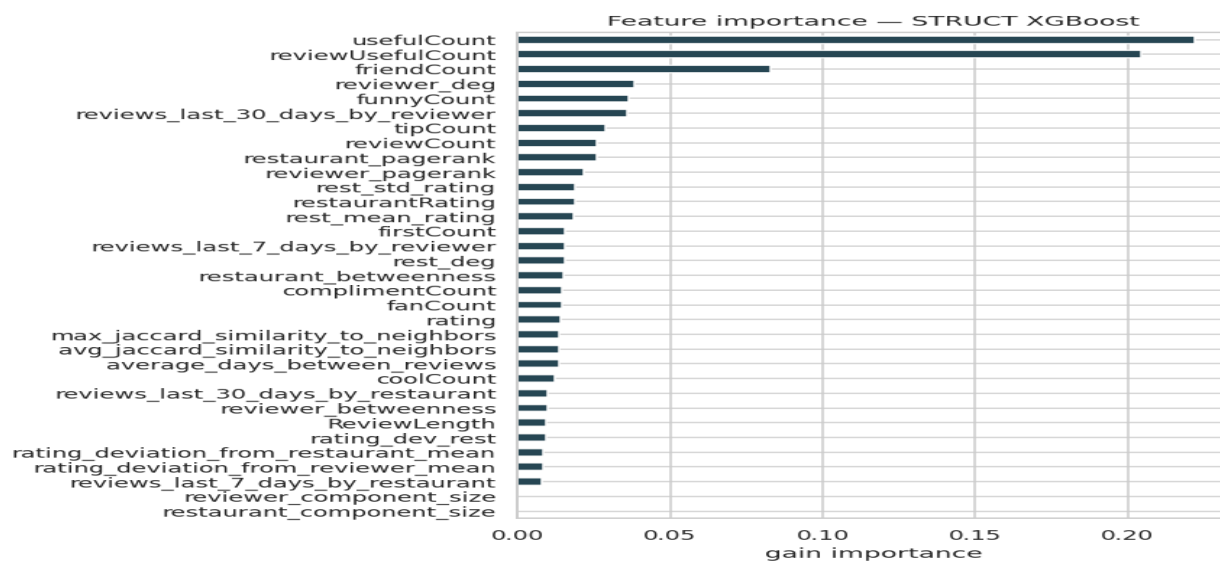


Figure 4: Feature importance — STRUCT x XGBoost

SECTION 7

Costs Incurred

Resource / Activity	Estimated Effort	Notes
Business Understanding (HW1)	4–6 hours	Objectives, success criteria, risk analysis
Data Understanding & EDA (HW2)	6–8 hours	EDA plots, leakage audit, data quality check
Data Preparation + Graph Construction	8–10 hours	Cleaning, train-only graph, 33-feature pipeline
Leakage correction (unplanned)	4–6 hours	Full rebuild of Section 5b after leakage found
Modeling — 3x3 factorial	3–4 hours wall time	Tree models on 5,000-dim sparse matrix: ~30-60s each
Hyperparameter tuning (RandomizedSearch)	1–2 hours	40 iterations x 5-fold CV on STRUCT features
Evaluation & report writing (HW3+HW4)	6–8 hours	All CRISP-DM deliverables
Presentation (23 slides)	3–4 hours	Full pipeline visual deck
Total estimated effort	~35-44 hours	Across all group members combined

Computational costs were minimal — all modeling was performed on CPU using scikit-learn and XGBoost. The most compute-intensive step was training tree models (RF, XGBoost) on the 5,000-dimensional sparse TF-IDF matrix (30–60 seconds per model on a standard laptop). No cloud compute, GPU, or paid services were required.

SECTION 8

Deviations from the Original Plan

The original project plan was followed closely. One significant unplanned event occurred:

Deviation	Original Plan	Actual Outcome	Impact
Leakage in graph features	Build graph on combined data; extract features; train model	Found that train+test graph inflated AUROC. Rebuilt entire Section 5b using TRAIN-only graph (Gtr), frozen lookup tables, and mapped features	4-6 extra hours. Final model is more realistic and production-ready
Hyperparameter tuning outcome	Expected tuning to meaningfully improve AUROC	RandomizedSearchCV gain < 0.002 — default config was already near-optimal	No change to the final model; confirms robustness

Text feature contribution	Expected HYBRID to outperform STRUCT	HYBRID approximately equals STRUCT for all classifiers; text adds no meaningful signal	Simplified deployment: only 33 structured features needed in production
GNN experiment (Section 9)	Optional GNN / Graph Autoencoder experiment planned	Not executed — structured features already dominated; GNN unlikely to add value given 88% degree-1 reviewers	No impact on business objectives; reduces complexity

SECTION 9

Implementation Plan (Deployment)

The approved model (STRUCT x XGBoost) is intended for deployment as a batch scoring service that ingests new review records daily, computes features, scores each review, and pushes a ranked list of high-risk reviews to a moderation dashboard.

Step	Description	Owner	Est. Effort
1	Serialize artifacts: save XGBoost, StandardScaler, and all TRAIN lookup tables as model_v1.pkl	ML Engineer	0.5 day
2	Probability calibration: fit isotonic regression; verify calibration curve before release	Data Scientist	1 day
3	Inference wrapper: score_reviews(df) function applying full feature pipeline + scoring	ML Engineer	1 day
4	REST API (FastAPI) or scheduled batch job (Airflow/cron) for daily scoring	MLOps Engineer	2-3 days
5	Moderation dashboard: surface risk_score, top features, low_confidence flag	Product / Frontend	3-5 days
6	Shadow mode: run in parallel with human moderation for 2-4 weeks before full rollout	Data Scientist + Ops	2-4 weeks

Cold-start handling: 73.8% of test reviewers are unseen at scoring time. Reviewer-level features fall back to global TRAIN means. A low_confidence flag is surfaced in the moderation UI for these cases.

Monitoring targets (weekly):

Metric	Target	Alert Threshold	Action
AUROC (labeled sample)	≥ 0.93	< 0.90	Trigger retraining investigation
Precision@top-5%	≥ 0.95	< 0.90	Adjust threshold; review false positives
False positive rate	$\leq 5\%$	$> 8\%$	Raise threshold; investigate feature drift

Score distribution (KS test)	$p > 0.05$	$p \leq 0.05$	Inspect new batches; possible concept drift
% cold-start reviews	Stable ± 5 pp	$> +15$ pp spike	Check for bot wave of new accounts

SECTION 10

Recommendations for Future Work

The following improvements are recommended for future iterations, ordered by expected value-to-effort ratio:

Priority	Recommendation	Expected Impact
High	Probability calibration (isotonic regression) — 1-day task	Risk scores become true probabilities; threshold decisions more reliable
High	Cold-start AUROC: evaluate separately on the 73.8% unseen-reviewer rows	Understand actual production performance for new users
Medium	Cross-reviewer collusion features: reviewer pairs that repeatedly target the same restaurants	Catch coordinated fake campaigns invisible to single-reviewer features
Medium	Quarterly retraining pipeline with automated holdout AUROC comparison	Maintain model performance as reviewer behavior evolves
Low	GNN experiment (GraphSAGE / GAT) on bipartite graph	May capture higher-order patterns; likely marginal gain given STRUCT dominance
Low	Text quality signals: incoherence, template-matching, repetition scores	May help on text-only cold-start reviewers with no behavioral history

SECTION 11

Conclusions

This project demonstrates that graph-based and behavioral features are dramatically more effective than text-based features for fake review detection on a Yelp-style platform. The key insight — that fake reviewers leave structural traces in the reviewer–restaurant interaction network — is validated decisively by the experimental results.

- The STRUCT x XGBoost model achieves AUROC = 0.949, PR-AUC = 0.937, and F1 = 0.860 on the held-out test set — meeting all six business success criteria.
- The 25.7% AUROC improvement over the text-only baseline confirms the project's central hypothesis.
- Feature engineering is the dominant factor: the choice of feature set explains far more variance in AUROC than the choice of classifier.

- The leakage detection and correction episode reinforces the importance of train-test isolation in graph feature engineering.
- The model is immediately deployable: it is fast, interpretable, and produces a high-quality moderation queue (Precision@top-5% = 0.984).
- The main production challenge is cold-start (73.8% of test reviewers unseen), which is manageable with fallback features and a low_confidence flag.

End of Final Report | Project Group 4 | Smeh Diab, Nir Levon, Shir Fox, Neta Nachteiler